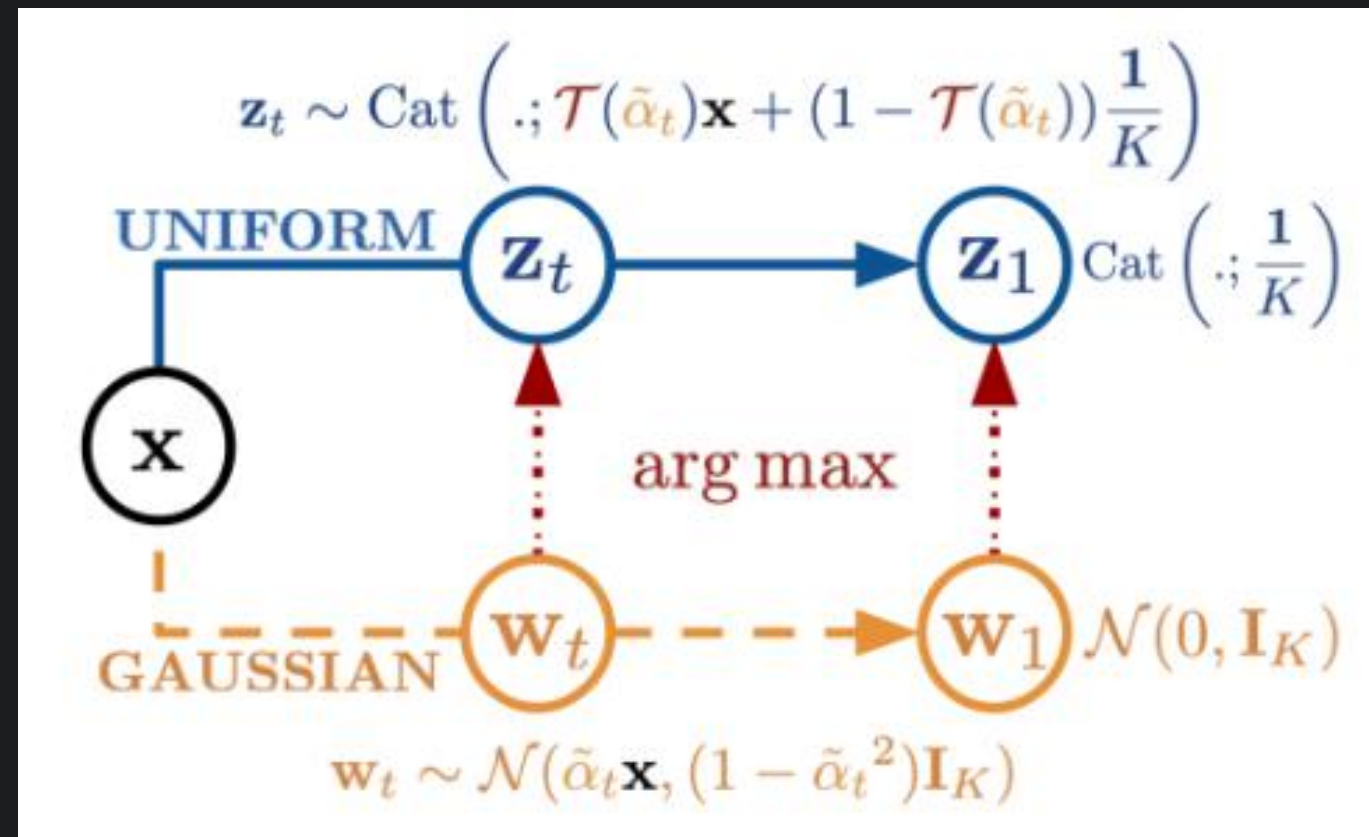


DUO Language Model Training

A comprehensive analysis of reduced-scale reproduction studies, loss-weighting functions, and frequency-based noise scheduling in transformer-based language modeling



Duo (Ours)

Sampling step: 8/8

Many years later, as he faced the firing squad, Colonel Aureliano Buendía was to remember that distant afternoon when his father took him to discover ice. At that time Macondo was a village of twenty adobe houses, built on the bank of a river of clear water that ran along a bed of polished stones, which were white and enormous, like prehistoric eggs. The world was so recent that many things lacked names, and in order to indicate them it was necessary to point.

Masked Diffusion

Sampling step: 30/30

Many years later, as he faced the firing squad, Colonel Aureliano Buendía was to remember that distant afternoon when his father took him to discover ice. At that time Macondo was a village of twenty adobe houses, built on the bank of a river of clear water that ran along a bed of polished stones, which were white and enormous, like prehistoric eggs. The world was so recent that many things lacked names, and in order to indicate them it was necessary to point.

AR

Sampling step: 37/84

Many years later, as he faced the firing squad, Colonel Aureliano Buendía was to remember that distant afternoon when his father took him to discover ice. At that time Macondo was a village of twenty adobe houses,

Research Methodology Overview

Reduced-Scale Reproduction

Systematic replication of DUO architecture at smaller scale to validate core training dynamics and architectural choices

Loss-Weighting Exploration

Comprehensive investigation of different loss-weighting functions to optimize training stability and convergence patterns

Frequency-Based Scheduling

Novel implementation of frequency-based noise scheduling techniques to improve model robustness and generalization capabilities

Loss Reweighting via Demystifying Diffusion Objectives

Standard NELBO Formulation

Masked discrete diffusion applies cross-entropy loss on masked tokens, scaled by a schedule-dependent coefficient:

$$\text{CE_weight} \propto \frac{\alpha'_t}{1-\alpha_t}$$

Demystifying Reinterpretation

Reweighted losses can be interpreted as weighted sums of tighter time-dependent ELBOs with general form:

$$\text{CE_weight} \propto \hat{w}(t) \cdot \frac{\alpha'_t}{1-\alpha_t}$$

where $\hat{w}(t)$ is monotone increasing with positive weights.

Weighting Strategies			
	Simple	Sigmoid	FM
$\hat{w}(t)$	$-\frac{1-a_t}{a'_t}$	$\frac{1-a_t}{1-(1-e^{-k})a_t}$	$\sqrt{\frac{1-a_t}{a_t}}$

Noise Schedule Design and Its Critical Role

MDLM Scheduling Framework

MDLM samples a masking rate $(1-\alpha_t)$ per training step; only masked tokens contribute to the weighted average over noise levels.

❏ For some schedules, soften or omit the derivative factor α'_t in the ELBO using power $p < 1$ or $p = 0$

Linear Schedule

Masking rates uniformly distributed with
 $E[1-\alpha_t] \approx 0.5$

Cosine Schedule

Concentrates on lower masking rates
with $E[1 - \alpha_t] \approx 0.36$, improving zero-
shot likelihood

Advanced Schedules

Unimodal/bimodal Gaussian
distributions over $(1-\alpha_t)$; bimodal can be
particularly effective

Datasets and Working Environment

Dataset Configuration

Training and Validation: Custom subset of OpenWebText containing 10,000 random documents

Zero-shot Evaluation Benchmarks:

- LM1B
- PTB
- WikiText
- LAMBADA

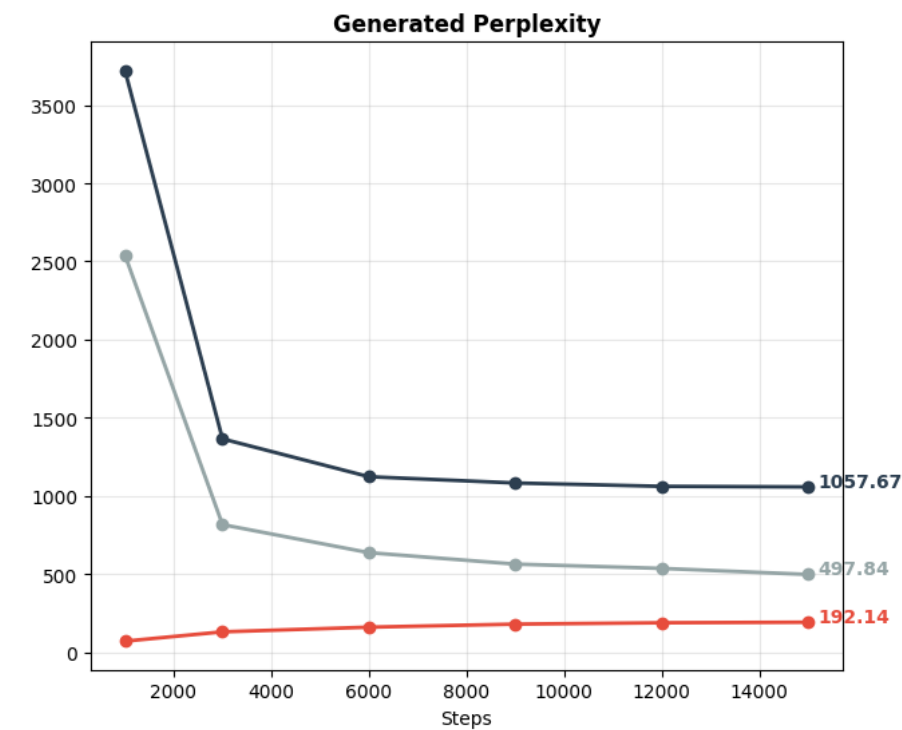
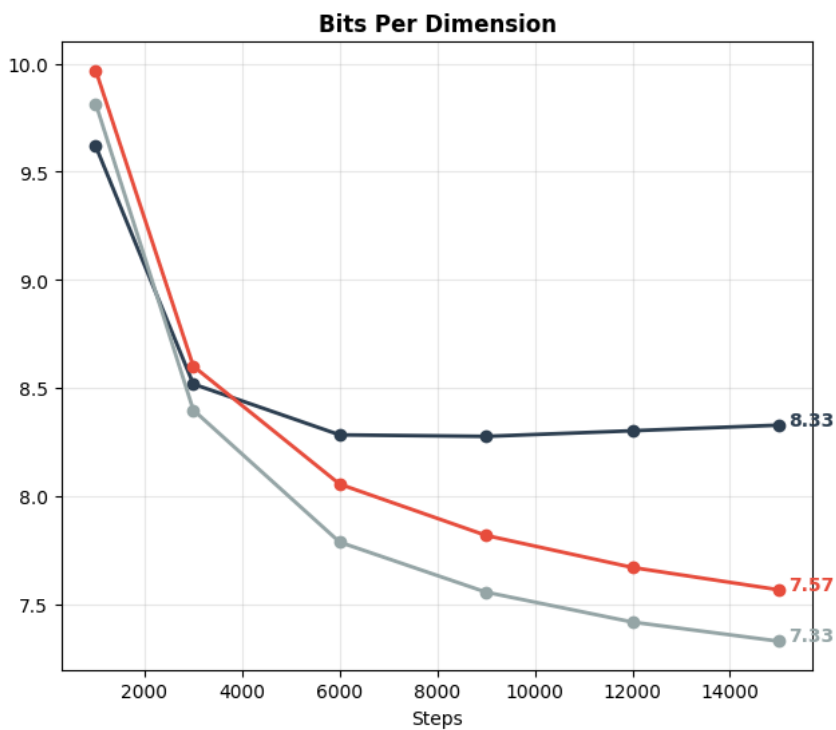
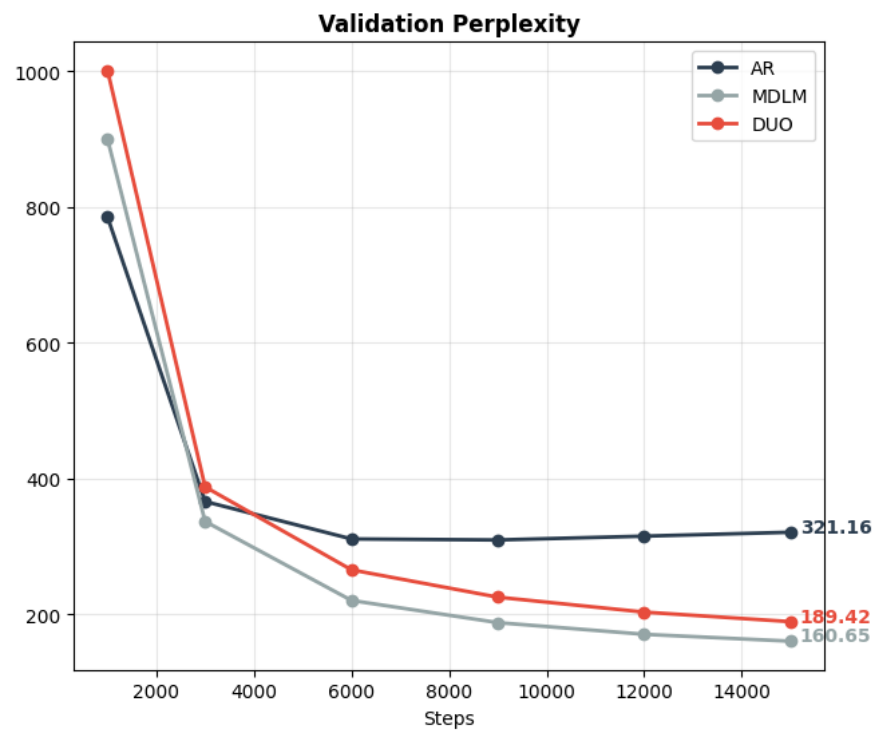
Computing Infrastructure

Primary Hardware: T4 and P100 GPUs via Kaggle's free tier, enabling accessible experimentation without extensive computational resources

Development Tools: VS Code for plot generation

Reproduction Results

Baseline Comparison (AR vs MDLM vs DUO)

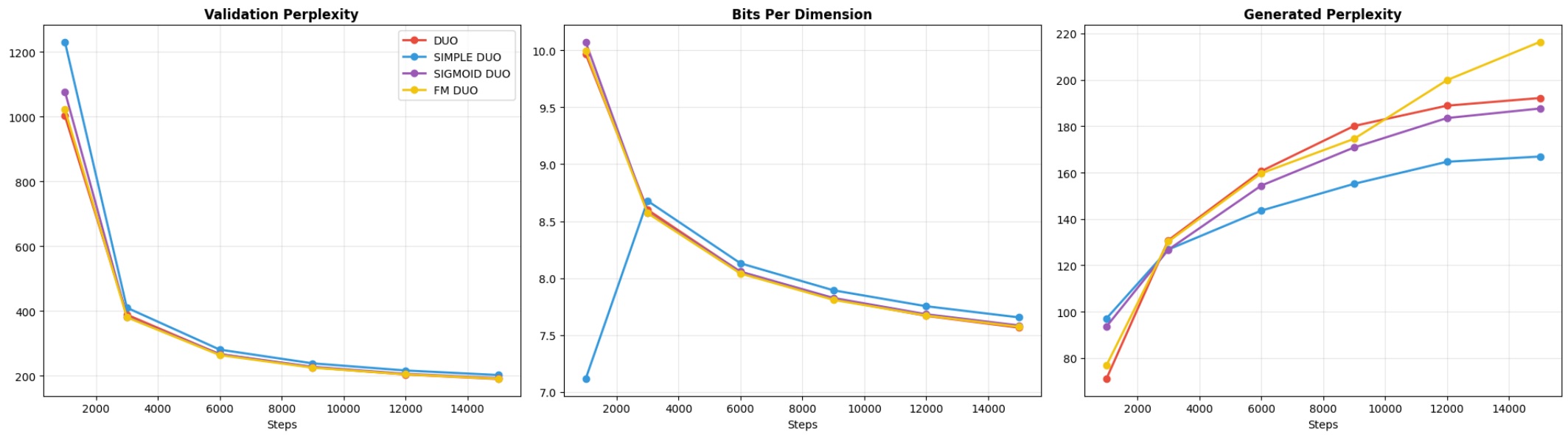


Reproduction Results (cont.)

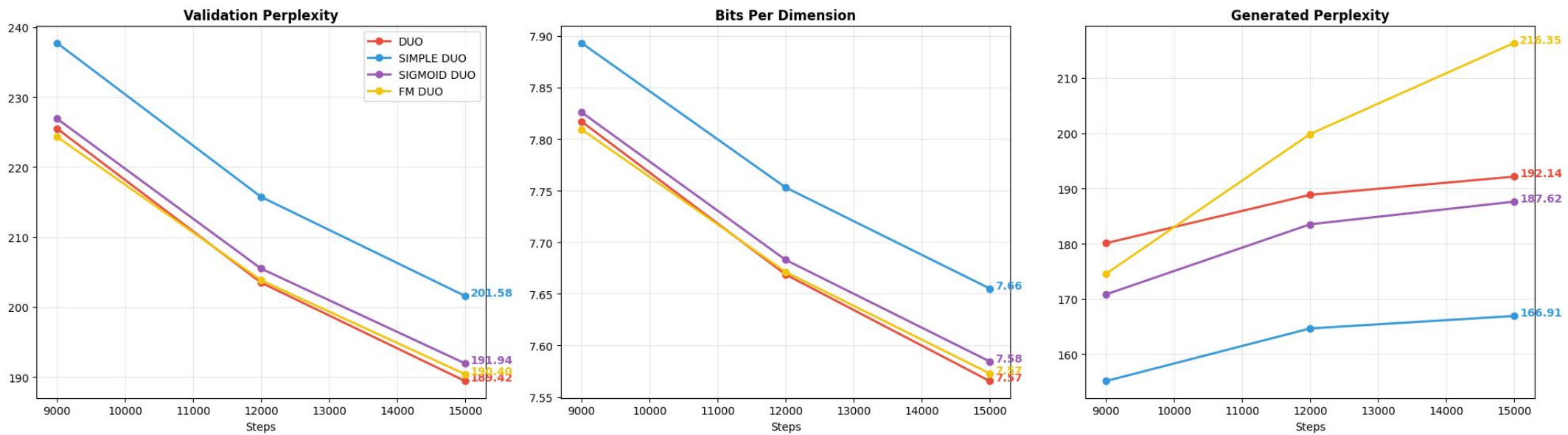
DUO's Generated Text per Number of Training Steps	
Steps	A Sample of the Generated Text
1000	The, of the way of the first, to the,, the way of the ways to be the same of,, in the you is the way it, and the lot of the lot of have the it.
3000	If you think this is the same thing. It is like. The it is not idea by the fact that many other people can be more more than more and get their work with your own.
6000	It is just many people as they have the other people to help them? Some of people who have access to other people.
9000	The president of the president said he will never do, the money the want to see his middle-class life. It's not at just a new years ago, the House said in a lot more about it.
12000	I can say this is the best game of the Week 2. If there is not surprise, the fact that being good and part of a new game, and then the team has different offensive issues.
15000	The FBI issued a report on this issue, but in favor of prosecution for Tuesday the activities, included have most important investigations and look.

Loss-reweighting Results

Weight Formula Innovation (DUO Variants)

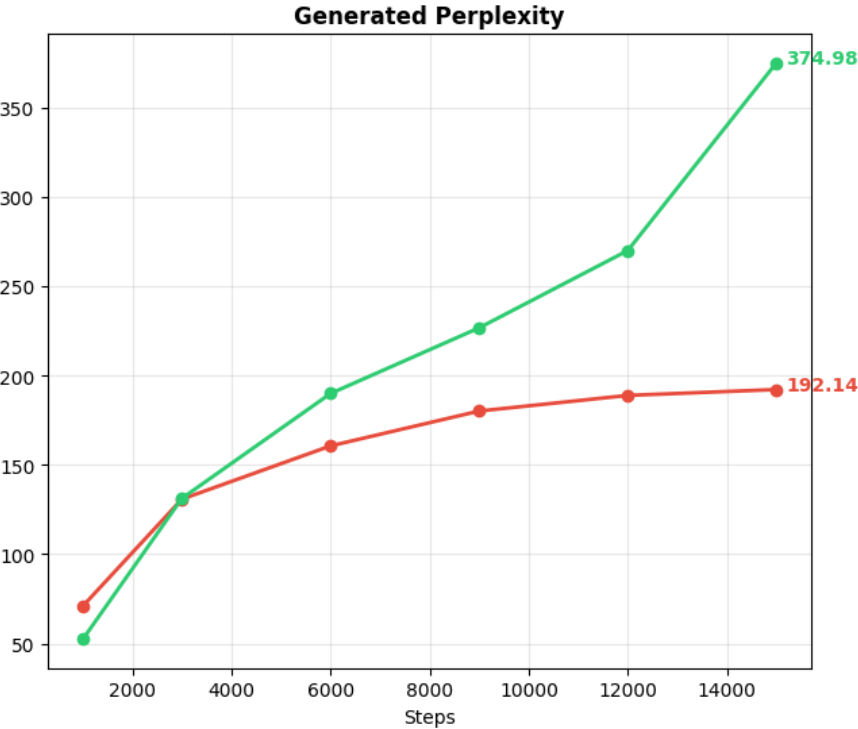
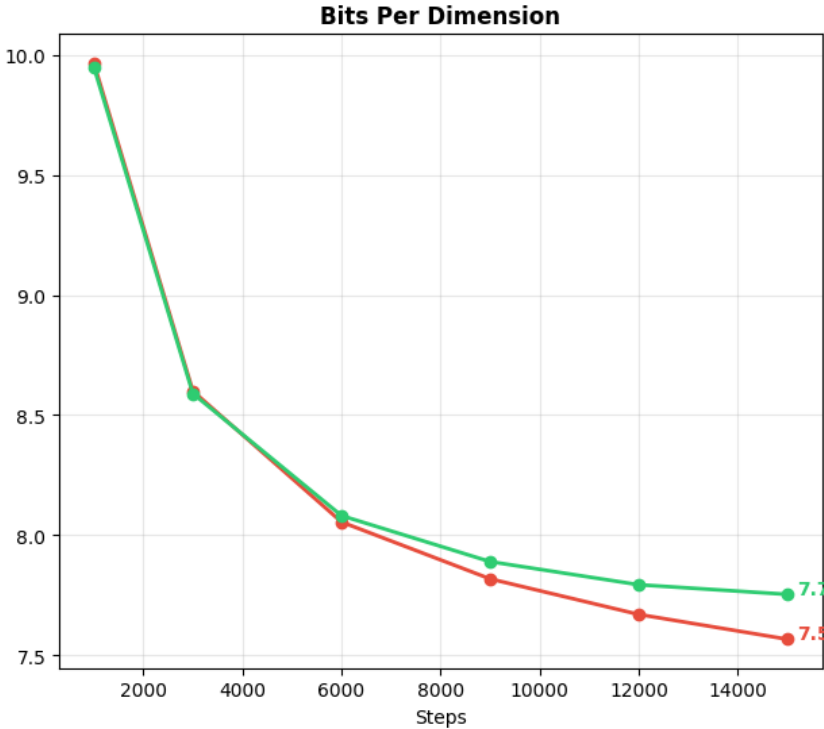
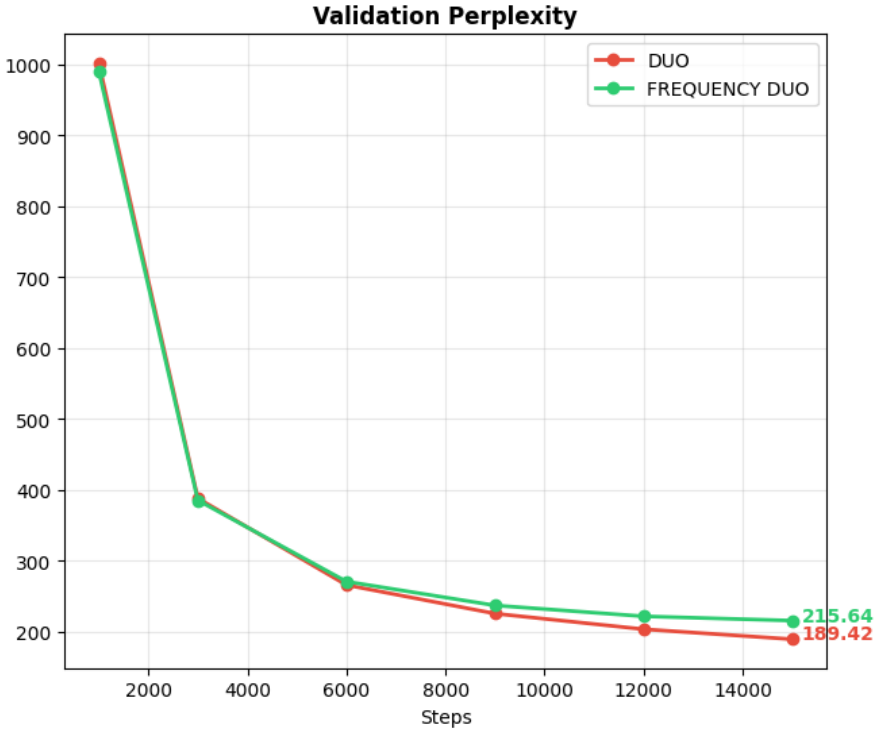


Convergence Phase Zoom (9k - 15k Steps)



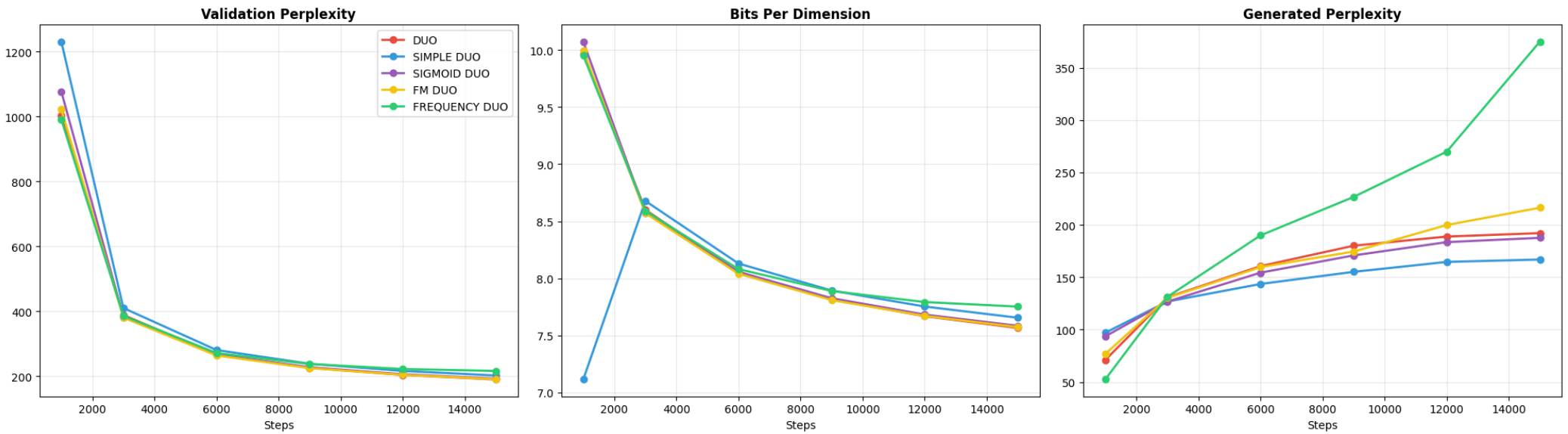
Frequency Informed Scheduling Results

Frequency Informed Innovation (DUO vs FREQUENCY DUO)

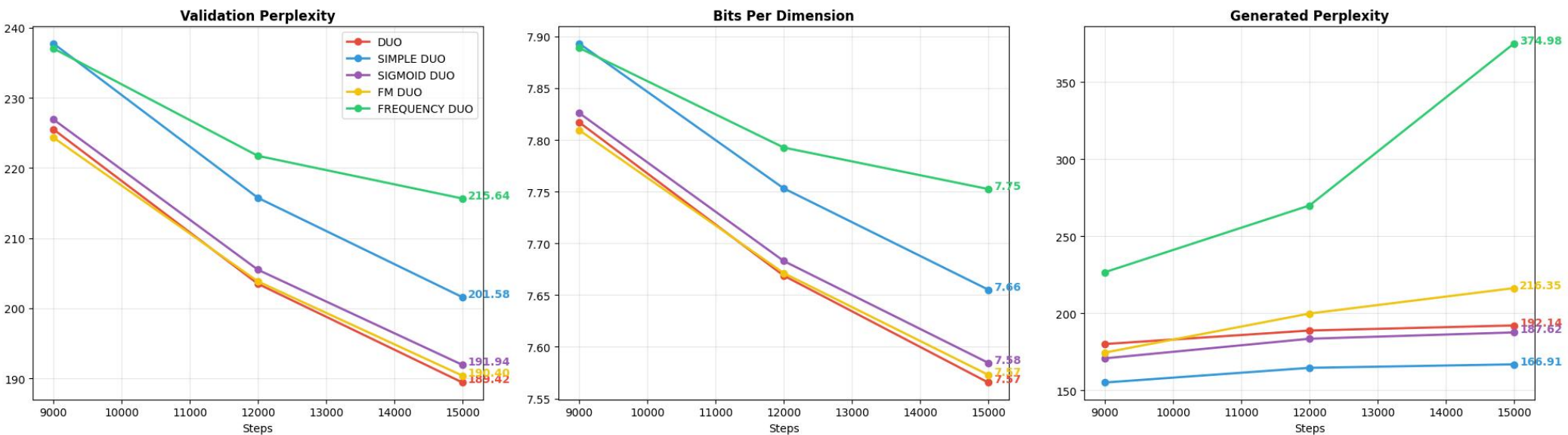


DUO Variants Results

Performance Across All DUO Modifications



DUO Family Convergence Zoom (9k - 15k Steps)

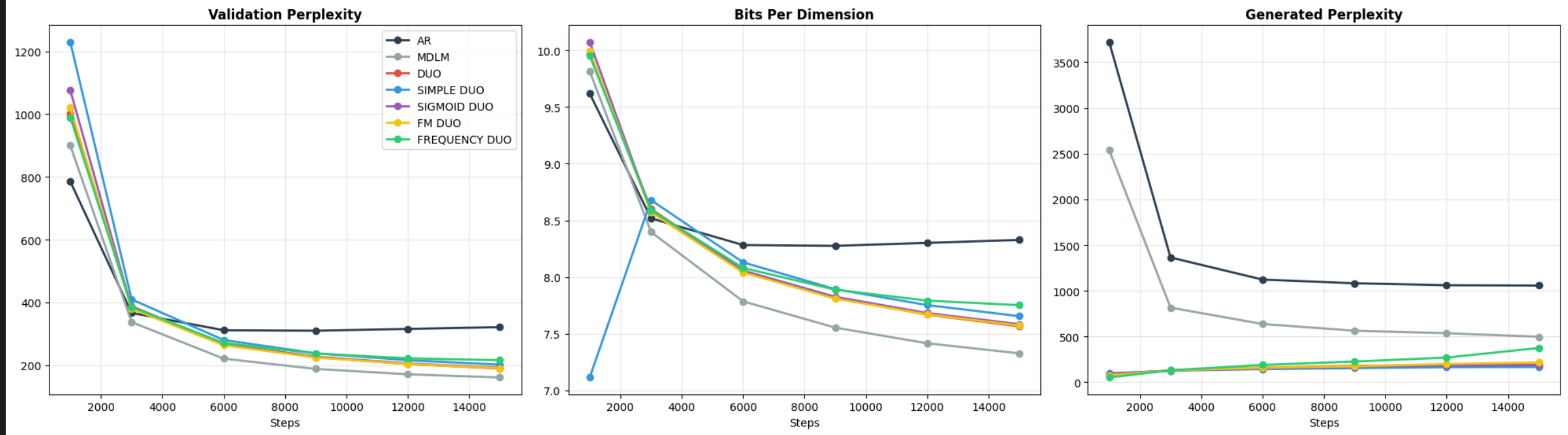


DUO Variants Results (cont.)

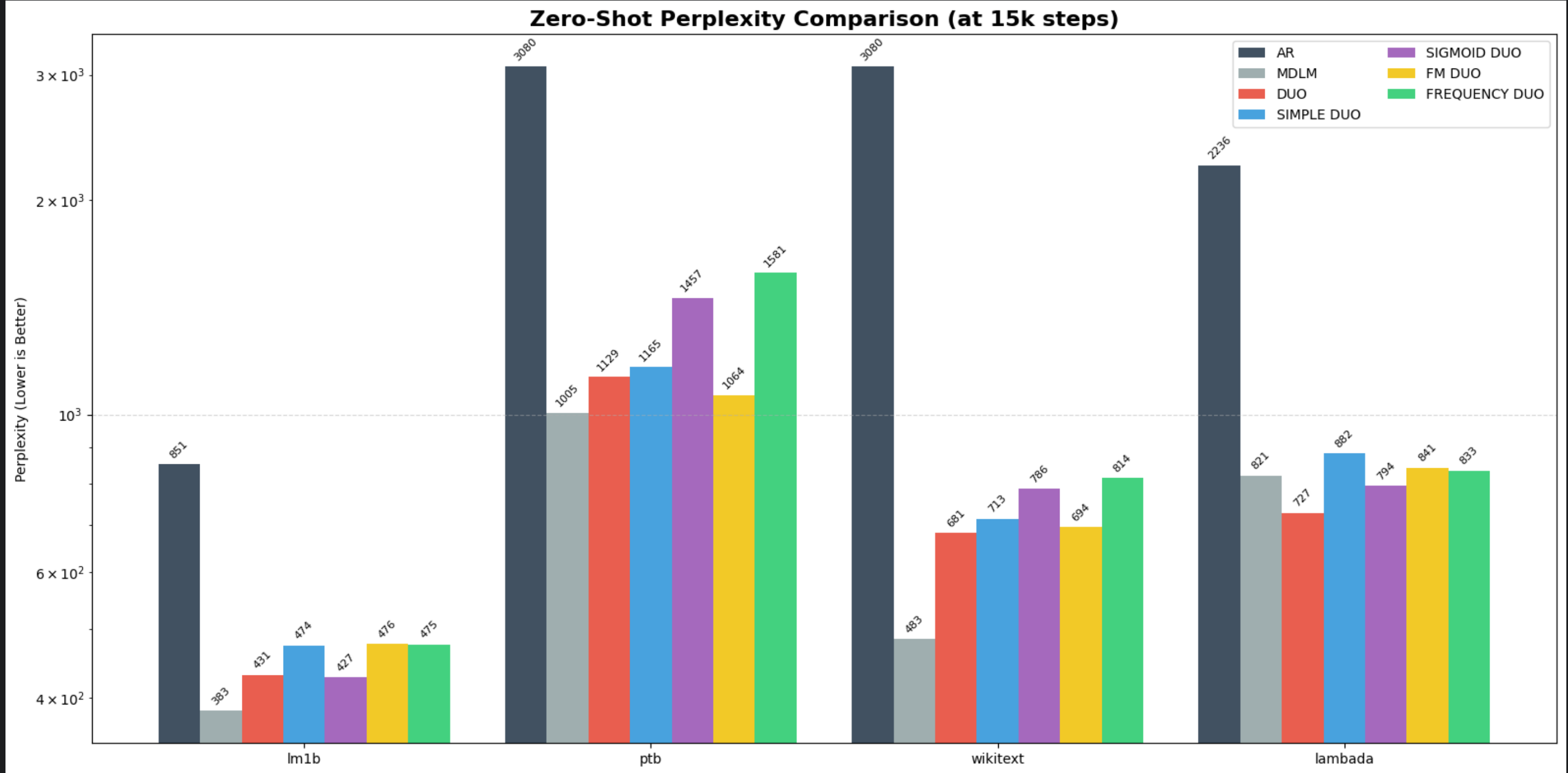
Generated Text for DUO Variants at 15000 Steps	
Model	A Sample of the Generated Text
SIGMOID DUO	Political areas with a national terrorist trafficking from other people. With the response to the inc demonization of Angela Merkel, and he returned to the country for two years. But he and his former campaign is the what he wanted to be all all.
FREQUENCY DUO	We think it it things the a the companies like the Lans Bailey and Big National Street Award. I am on board with freelance writer. Van and Ing Holaer and Eroval, who been several back at the Boston Film Library.

All Models Results

All 7 Experimental Models



All Models Results (cont.)



Task Distribution

Name	Task
Spyridon Agathos	Theoretical Comprehension and Paper Research
Konstantinos Leivadas	
Stefanos Rompos	Paper Reproduction and Technical Implementations
Kris Koutsi	

Future Work

- Bigger Training Dataset and Diffusion Model
- More Computational Power
- Alternative Noise Scheduling

Thank You!

Q & A